Stereo Vision

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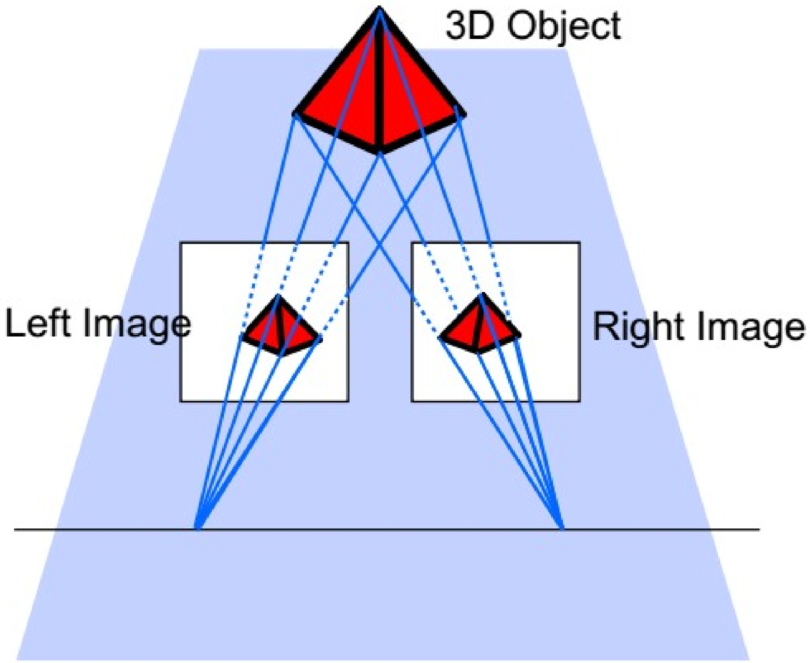
3.1 Stereo Vision

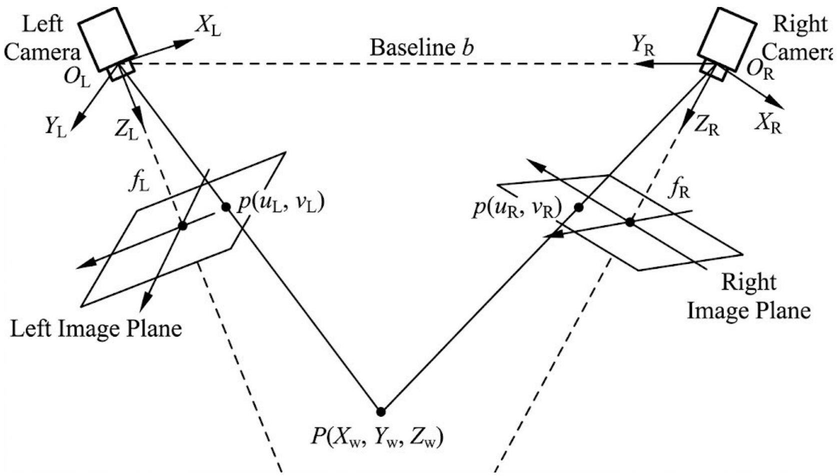
- Stereo Vision is the process of obtaining depth information from a pair of images captured by two cameras that observe the same scene from different but known positions.
- Stereo Vision is widely used in fields such as 3D navigation, 3D positioning, object tracking, and robotics research.
- Structure from Motion is the process of obtaining depth and motion information from a pair of images captured by the same camera that views the same scene from different positions.

3.1 Stereo Vision

- In the world coordinate system, any point $\boldsymbol{P}(X_w, Y_w, Z_w)$ on the surface of the measured object is projected onto the image plane of the left camera coordinate system as $p_L(u_L, v_L)$ and onto the image plane of the right camera coordinate system as $p_R(u_R, v_R)$.
- Using a binocular vision system with two cameras, if the image points (u_L, v_L) and (u_R, v_R) can be confirmed to correspond to the same spatial feature point $\boldsymbol{P}(X_w, Y_w, Z_w)$, the exact position of point $\boldsymbol{P}(X_w, Y_w, Z_w)$ can be calculated from the intersection of the two lines formed by connecting the two image points to the optical centers of the respective cameras.



3.1 Stereo Vision



A binocular vision system can reconstruct three-dimensional spatial information based on the distance between two homologous points in the images captured by two cameras using trigonometric geometry. Homologous points are the image points of the same spatial point in different images, also known as conjugate points. The difference in their positions within the images is called disparity.

3.1 Stereo Vision

When studying binocular systems, it is generally necessary to first clarify the following types of coordinate systems involved in the research process:

1. World coordinate system. A pre-selected reference 3D coordinate system in the environment used to describe the coordinate positions of the camera and objects in the environment. In a binocular vision system, the left camera coordinate system is generally defined as the world coordinate system.
2. Camera coordinate system. A 3D coordinate system with the camera's optical center as the origin and the optical axis as the Z-axis. In a binocular system, the left and right camera coordinate systems can be transformed through rotation and translation.
3. Image coordinate system. The 2D coordinate system of the image formed by the camera, which can be described by pixel count or physical dimensions. For a 3D spatial point in the camera coordinate system, the position of its image point in the image coordinate system can be calculated through triangular projection transformation.

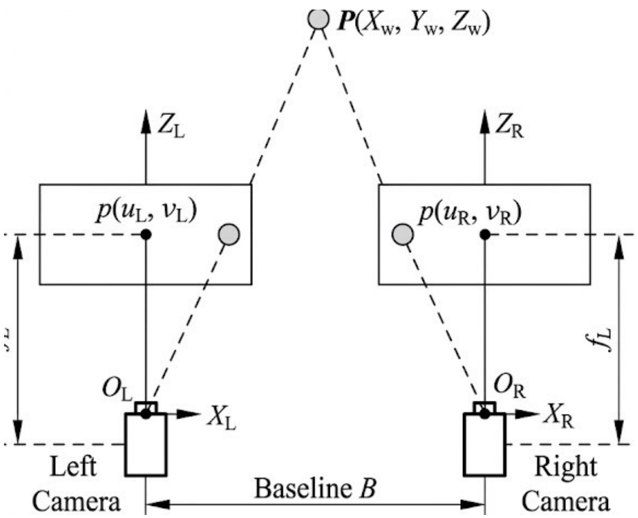
3.1 Stereo Vision

- If the left camera coordinate system is defined as the world coordinate system, then the position of point $\boldsymbol{P}_w(X_w, Y_w, Z_w)$ in the world coordinate system projected onto the image coordinate system of the left camera can be directly calculated through the perspective projection relationship.
- The position of this point in the right camera's image coordinate system can also be obtained through projection transformation calculation.

$$Z_R \begin{bmatrix} u_R \\ v_R \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_R & 0 & 0 & 0 \\ 0 & f_R & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{投影变换}} \underbrace{\begin{bmatrix} \boldsymbol{R} & \boldsymbol{B} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}}_{\boldsymbol{P}_R(X_R,\ Y_R,\ Z_R)}$$

$$Z_w \begin{bmatrix} u_L \\ v_L \\ 1 \end{bmatrix} = \begin{bmatrix} f_L & 0 & 0 & 0 \\ 0 & f_L & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

3.1 Stereo Vision: Parallel Binocular System



Characteristics of Parallel Binocular Vision System:

(1) The two camera types are identical, o at least have the same focal length.

(1) The two cameras are installed in

(2) The two cameras are installed. If it's a single moving camera system the motion trajectory is a straight line.

(3) The X-axes of the two cameras are collinear and parallel to the baseline.

(4) The left camera coordinate system is defined as the world coordinate system.

An ideal simplified scenario assumes both cameras are identical and aligned with the X-axis. Can we derive an expression for the depth Z_w of point $\boldsymbol{P}_w$?

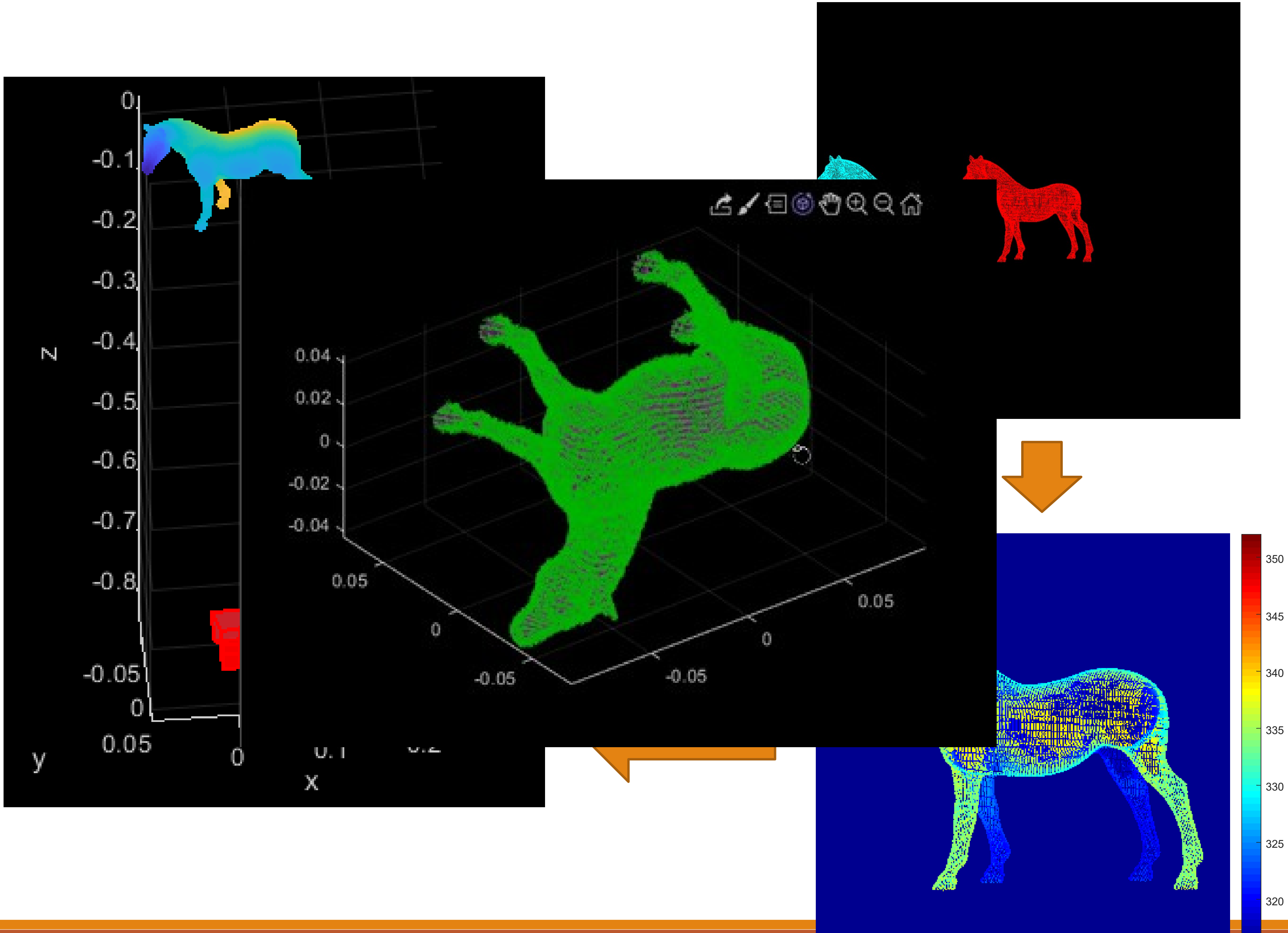
$$\begin{cases} u_L = f \cdot \dfrac{X_w}{Z_w} \\ u_R = f \cdot \dfrac{X_w - T_x}{Z_w} \\ v_L = v_R = f \cdot \dfrac{Y_w}{Z_w} \end{cases} \Rightarrow Z_w = f \cdot \frac{T_x}{d} = f \cdot \frac{T_x}{u_L - u_R}$$

3.1 Stereo Vision: Parallel Binocular System

- As long as any point on the left camera image plane can find a homologous matching point on the right camera image plane, the three-dimensional coordinates of that point can be determined based on the parallax.
- Furthermore, as long as all points on the image plane of the measured target can find homologous points in the other camera, the three-dimensional coordinates of the measured target can be determined.

$$Z_{\mathrm{w}} = f \cdot \frac{T_x}{d} = f \cdot \frac{T_x}{u_{\mathrm{L}} - u_{\mathrm{R}}}$$

$$d = u_{\mathrm{L}} - u_{\mathrm{R}} = f \cdot \frac{T_x}{Z_{\mathrm{w}}}$$


0
-0.1
-0.2
-0.3
-0.4
-0.5
-0.6
-0.7
-0.8
z
-0.05
0
0.05
y
0
x
0.04
0.02
0
-0.02
-0.04
0.05
0
-0.05
-0.05
0
0.05
350
345
340
335
330
325
320

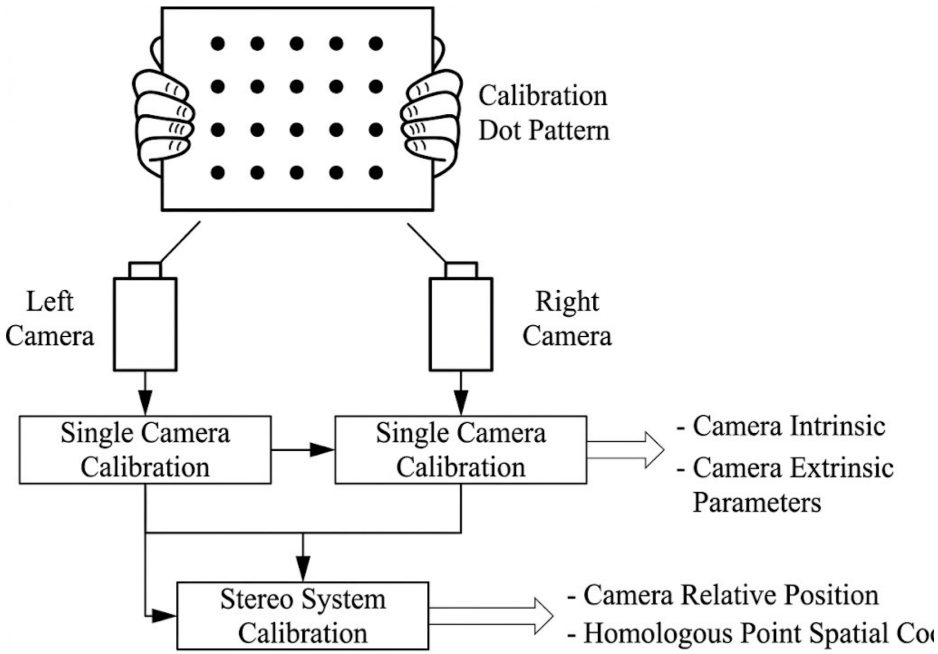
3.2 Stereo Vision System Calibration

To calculate three-dimensional depth information using the disparity of a stereo vision system, the following information about the stereo vision system must first be obtained:

1. The internal parameters of the cameras. This includes the focal length, optical center, and distortion model of each camera.
2. The external parameters of the cameras. This includes the rotation and translation relationships between each camera coordinate system and the world coordinate system.
3. The relative positional relationships between the cameras. This includes the rotation and translation relationships between each camera coordinate system and other camera coordinate systems, as well as the coordinates of spatial points corresponding to homologous image points in multiple images.

3.2 Stereo Vision System Calibration

- In actual engineering practice, the process of obtaining the aforementioned various information is referred to as the stereo calibration process of a stereo vision system.
- To achieve the calibration of a stereo vision system, it is common to have each camera of the system capture images of the same calibration point pattern, and then calculate various parameters based on the point pattern information.



3.2 Stereo Vision System Calibration

- In a binocular stereo vision system, let the rotation and translation matrices of the left and right camera coordinate systems relative to the world coordinate system be ($\boldsymbol{R}_L$, $\boldsymbol{t}_L$) and ($\boldsymbol{R}_R$, $\boldsymbol{t}_R$) respectively. Then the coordinates of a point $\boldsymbol{P}_w(X_w, Y_w, Z_w)$ in the 3D world coordinate system in the two camera coordinate systems, $\boldsymbol{P}_L(X_L, Y_L, Z_L)$ and $\boldsymbol{P}_R(X_R, Y_R, Z_R)$, can be obtained through coordinate rotation and translation. This is expressed in matrix form as

$$\boldsymbol{P}_L = \boldsymbol{R}_L \boldsymbol{P}_w + \boldsymbol{t}_L$$

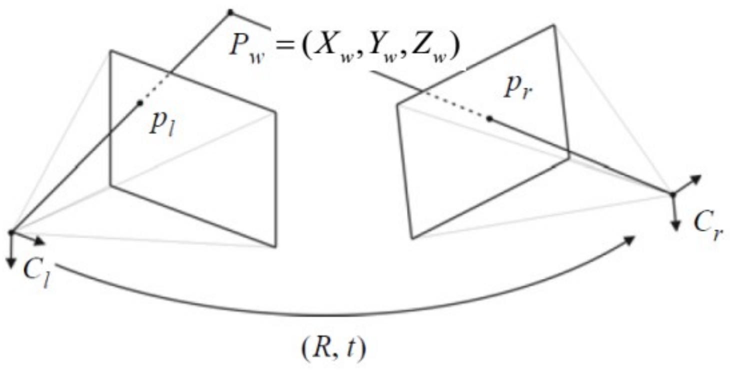
$$\boldsymbol{P}_R = \boldsymbol{R}_R \boldsymbol{P}_w + \boldsymbol{t}_R$$

$$\boldsymbol{P}_w = \boldsymbol{R}_L^{\top} \boldsymbol{P}_L - \boldsymbol{R}_L^{\top} \boldsymbol{t}_L$$

$$\boldsymbol{P}_R = \boldsymbol{R}_R \boldsymbol{R}_L^{\top} \boldsymbol{P}_L + \boldsymbol{t}_R - \boldsymbol{R}_R \boldsymbol{R}_L^{\top} \boldsymbol{t}_L$$

$$\boldsymbol{R}_{RL} = \boldsymbol{R}_R \boldsymbol{R}_L^{\top}, \quad \boldsymbol{t}_{RL} = \boldsymbol{t}_R - \boldsymbol{R}_R \boldsymbol{R}_L^{\top} \boldsymbol{t}_L$$

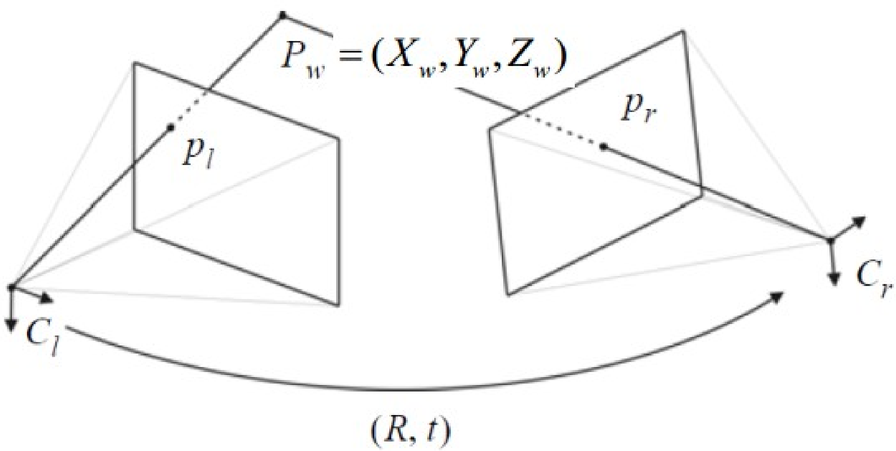
3.3 Stereo Vision | General Case



- In nature, no two cameras are identical, as small alignment errors always exist between them due to the manufacturing process.
- In practice, the exact location of a camera's optical center cannot be known precisely.
- It is extremely difficult to align two cameras on a horizontal axis. To use a stereo camera, it is necessary to know the intrinsic and extrinsic parameters of each camera, i.e., the relative pose (rotation, translation) between the cameras → this problem can be solved through camera calibration.

3.3 Stereo Vision | General Case

- To estimate the 3D position of $\boldsymbol{P}_w$, a system of equations can be constructed from the left and right cameras.
- Triangulation is the problem of determining the 3D location of a point given a set of corresponding image positions and known camera poses.



Left camera: set the world frame to coincide with the left camera frame

$$\tilde{p}_l = \lambda_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = K_l \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix}$$

Right camera:

$$\tilde{p}_r = \lambda_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = K_r R \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + T$$

3.3 Stereo Vision | Corresponding Search

3.3 Stereo Vision | Corresponding Search

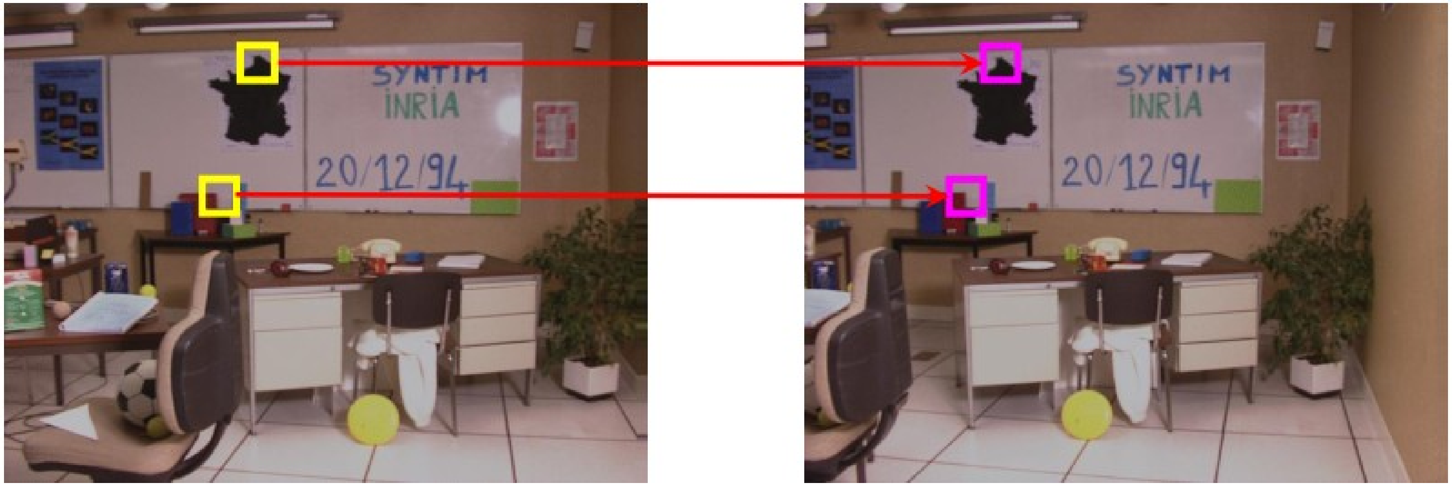
- Objective: Identify corresponding points in the left and right images, which are reprojections of the same 3D scene point
- Typical similarity measures: Normalized Cross-Correlation (NCC), Sum of Squared Differences (SSD), and consistency transformations

$$NCC = \frac{\sum_{k=-a}^{a}\sum_{l=-b}^{b}\left[I_1(u+k,v+l)-\mu_1\right]^2 \bullet \left[I_2(u'+k,v'+l)-\mu_2\right]^2}{\sqrt{\sum_{k=-a}^{a}\sum_{l=-b}^{b}\left[I_1(u+k,v+l)-\mu_1\right]^2 \sum_{k=-a}^{a}\sum_{l=-b}^{b}\left[I_2(u'+k,v'+l)-\mu_2\right]^2}}$$

$$\mathrm{SSD} = \sum_{k=-a}^{a}\sum_{l=-b}^{b}\left[I_1(u+k,v+l)-I_2(u'+k,v'+l)\right]^2$$

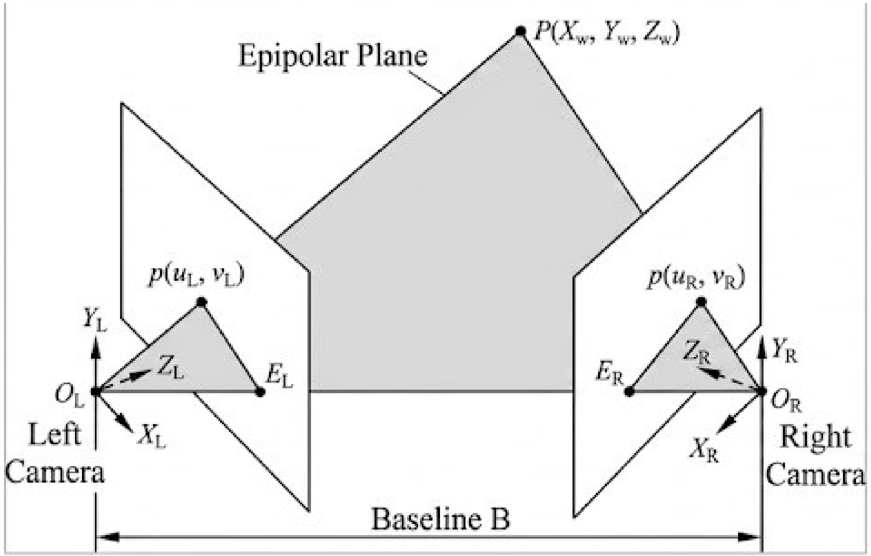
3.3 Stereo Vision | Corresponding Search

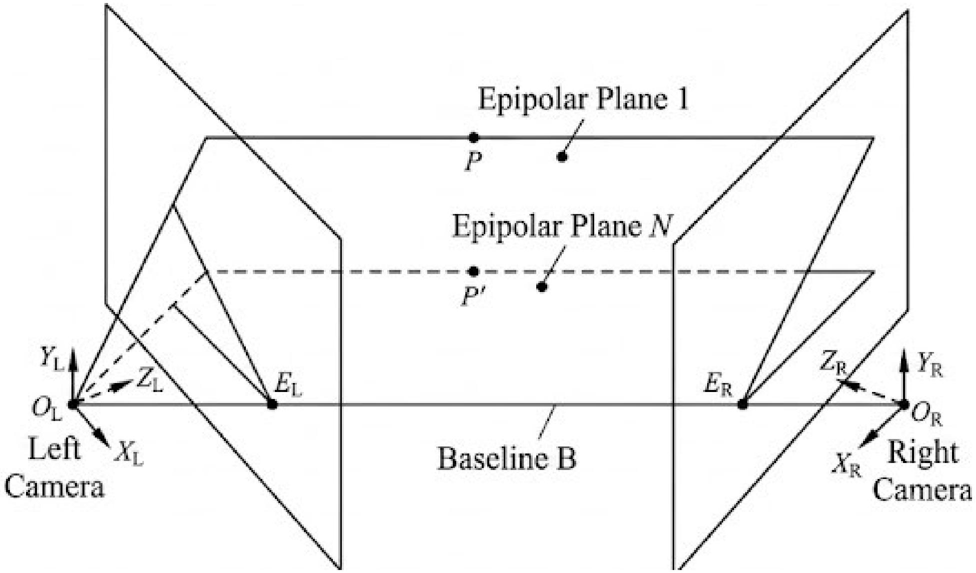
- Searching in two-dimensional images is computationally very expensive! Can corresponding searches be conducted in one dimension?



3.3 Stereo Vision | Epipolar Geometry

- Epipolar geometry theory can be used to study the relationships between homologous image points in the images captured by various cameras within a stereo vision system. It not only plays a significant role in matching corresponding points in binocular stereo vision images but also has broad applications in 3D reconstruction and motion analysis. The epipolar geometry in stereo vision systems typically involves the following key definitions:





3.3 Stereo Vision | Epipolar Geometry

- The vector cross product takes two vectors and returns a third vector perpendicular to both inputs.
- The cross product of two parallel vectors equals zero.
- The vector cross product can also be expressed as the product of a skew-symmetric matrix and a vector.

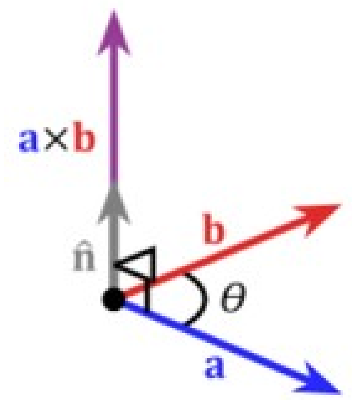
$$\vec{a} \times \vec{b} = \vec{c}$$

$$\|\vec{c}\| = \|\vec{a}\|\|\vec{b}\|\sin(\theta) \cdot \vec{n}$$

$$\vec{a} \cdot \vec{c} = 0$$

$$\vec{b} \cdot \vec{c} = 0$$

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = [\mathbf{a}_\times]\mathbf{b}$$



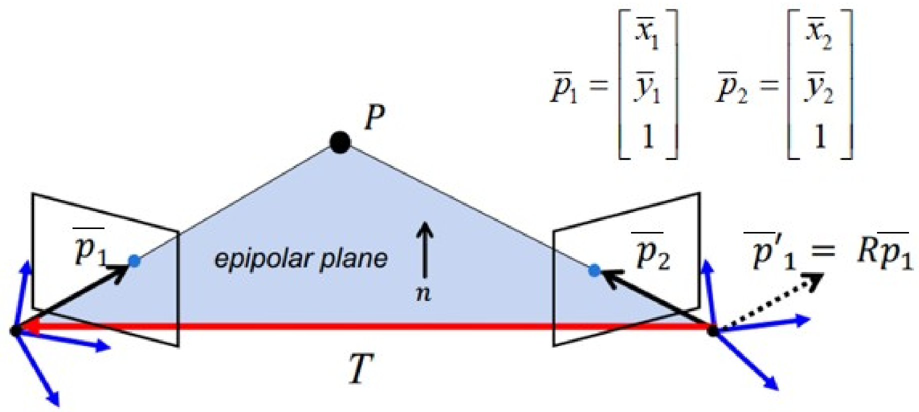
3.3 Stereo Vision | Epipolar Geometry

- $\overline{\boldsymbol{p}}_1$, $\overline{\boldsymbol{p}}_2$ and $\boldsymbol{T}$ are coplanar

$$\overline{\boldsymbol{p}}_2 \cdot (\boldsymbol{T} \times \overline{\boldsymbol{p}}'_1) = 0$$

$$\overline{p}'_1 = \boldsymbol{R}\overline{\boldsymbol{p}}_1$$

$$\overline{\boldsymbol{p}}_2^T [\boldsymbol{T}]_\times \boldsymbol{R} \overline{\boldsymbol{p}}_1 = 0$$



$$\overline{\boldsymbol{p}}_2^T \boldsymbol{E} \overline{\boldsymbol{p}}_1 = 0$$

Epipolar constraint

$$\boldsymbol{E} = [\boldsymbol{T}]_\times \boldsymbol{R}$$

Essential matrix

3.3 Stereo Vision | Epipolar Geometry

- The epipolar constraint equation reflects the constraint relationship between two homologous image points $\boldsymbol{p}_1(x_1, y_1)$ and $\boldsymbol{p}_2(x_2, y_2)$ in the camera coordinate systems.
- The essential matrix $\boldsymbol{E}$ is a 3×3 matrix with a rank of 2.
- The essential matrix actually describes the rotation and translation relationship between the left and right camera coordinate systems. It is only related to the relative positions between the cameras and is independent of the internal parameters of each camera, such as focal length, optical center, and sensor pixel size.
- The essential matrix is only applicable to the physical coordinate transformation of homologous image points in the two camera coordinate systems and cannot establish a connection between their image pixel coordinates.

3.3 Stereo Vision | Epipolar Geometry

$$\overline{\boldsymbol{p}}_2^T \boldsymbol{E} \overline{\boldsymbol{p}}_1 = 0 \qquad \left(\boldsymbol{K}_2^{-1}\overline{\boldsymbol{m}}_2\right)^T \boldsymbol{E}\left(\boldsymbol{K}_1^{-1}\overline{\boldsymbol{m}}_1\right) = 0 \qquad \overline{\boldsymbol{m}}_2^T\left(\boldsymbol{K}_2^{-T}\boldsymbol{E}\boldsymbol{K}_1^{-1}\right)\overline{\boldsymbol{m}}_1 = 0$$

$$\boldsymbol{F} = \boldsymbol{K}_2^{-T}\boldsymbol{E}\boldsymbol{K}_1^{T} \qquad \overline{\boldsymbol{m}}_2^T \boldsymbol{F} \overline{\boldsymbol{m}}_1 = 0$$

Fundamental Matrix **Epipolar constraint**

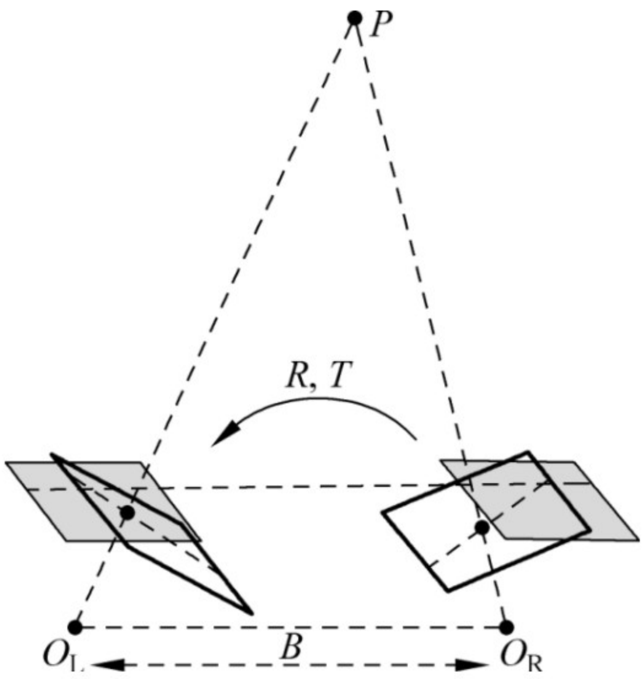
- The epipolar constraint can be expressed using the fundamental matrix through the following formulas:

  The right epipolar line corresponding to the left image point: $\tilde{\boldsymbol{l}}_R = \boldsymbol{F}\overline{m}_1$

  The left epipolar line corresponding to the right image point: $\tilde{\boldsymbol{l}}_l = \boldsymbol{F}^T\overline{m}_2$

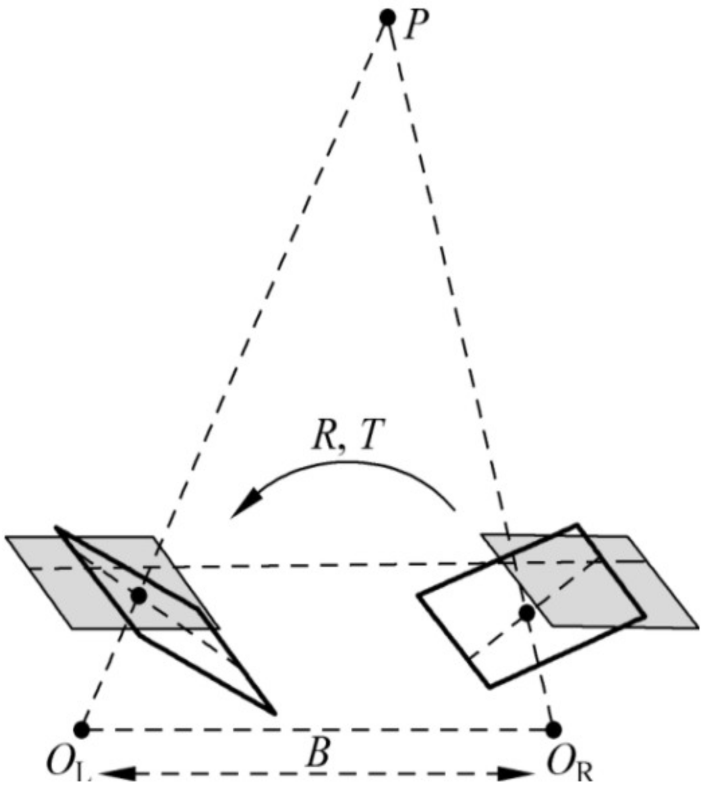
3.3 Stereo Vision | Epipolar Geometry

3.3 Stereo Vision | Image Adjustment

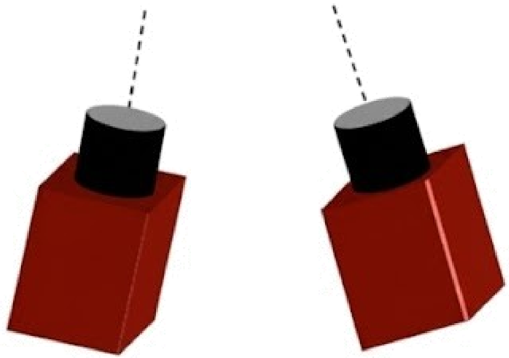


- Through the calibration process of single-camera and stereo vision systems, the relationship between the coordinates of homologous image points in different camera coordinate systems or image coordinate systems can be obtained.
- To calculate the depth information of spatial points based on disparity, further image adjustment (Image Rectification) is required to make the images from both cameras coplanar and "row-aligned" to simplify the calculation process.

3.3 Stereo Vision | Image Adjustment

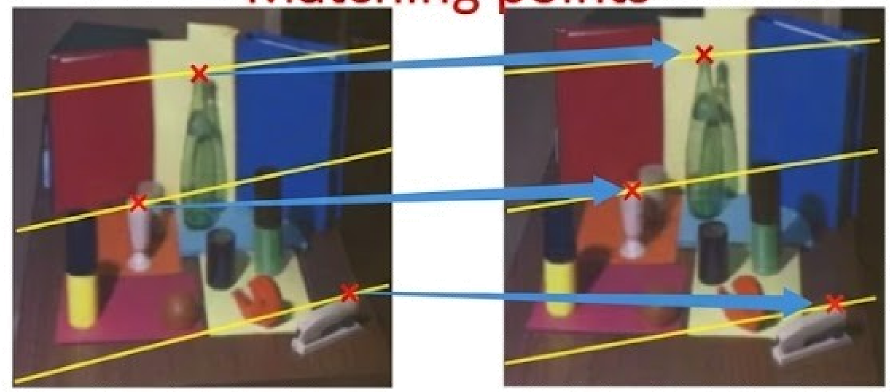


- Although the parallel binocular system model can greatly simplify the calculation of depth information, it is an extremely idealized model. In the real world, it is almost impossible to make the image planes of two cameras completely coplanar; therefore, a physically perfect parallel binocular system cannot be achieved, and one can only approximate its configuration as closely as possible based on its characteristics.
- However, through adjustment processes such as distortion correction, rotation, translation, and row alignment of the images, an "approximate fronto-parallel binocular system" can be transformed into a nearly ideal fronto-parallel binocular system.

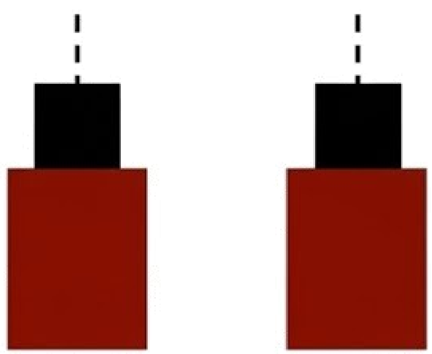


a. Before rectification

Matching points

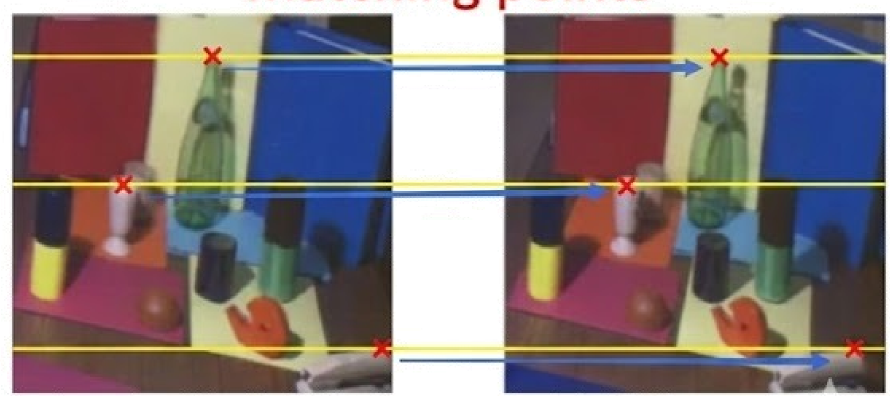


Original stereo pair



b. After rectification

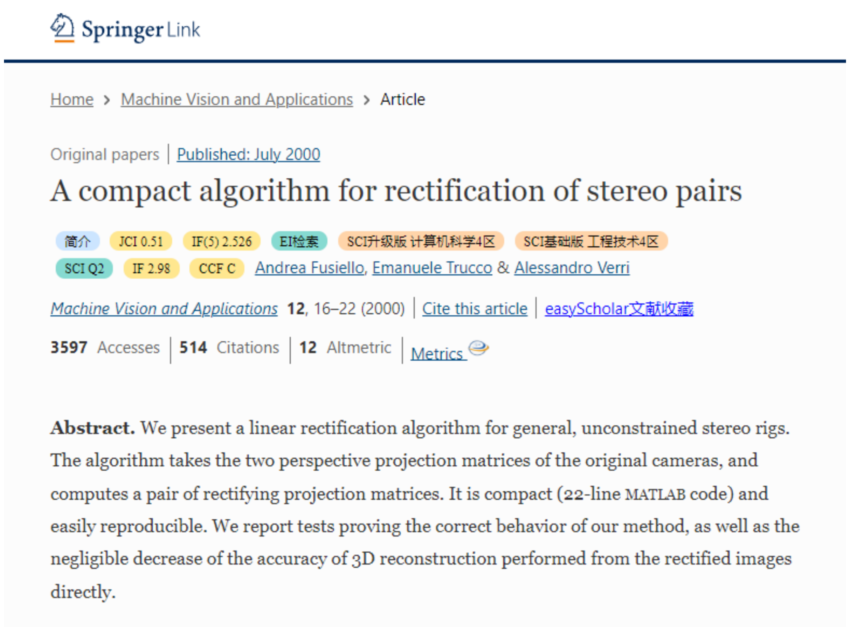
Matching points



Stereo pair in standard form

3.3 Stereo Vision | Image Adjustment

- Assume that the positional relationship between the left and right camera images can be represented by a rotation matrix $\boldsymbol{R}$ and a translation vector $\boldsymbol{T}$. Rotate the left and right image coordinates by $\boldsymbol{R}_L$ and $\boldsymbol{R}_R$ respectively, so that both become parallel to the sum of the vectors formed by the optical centers and projection centers of the two cameras (which is also a vector).
- Although rotating the left and right camera coordinates separately can make the two images coplanar, the pixel rows in the two coplanar images are not necessarily aligned. This is unfavorable for achieving the subsequent image alignment process, so further adjustments to the two images are required.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Original papers | Published: July 2000

A compact algorithm for rectification of stereo pairs

简介 JCI 0.51 IF(5) 2.526 EI检索 SCI升级版 计算机科学4区 SCI基础版 工程技术4区 SCI Q2 IF 2.98 CCF C Andrea Fusiello, Emanuele Trucco & Alessandro Verri

Machine Vision and Applications **12**, 16–22 (2000) | Cite this article | easyScholar文献收藏

3597 Accesses | **514** Citations | **12** Altmetric | Metrics

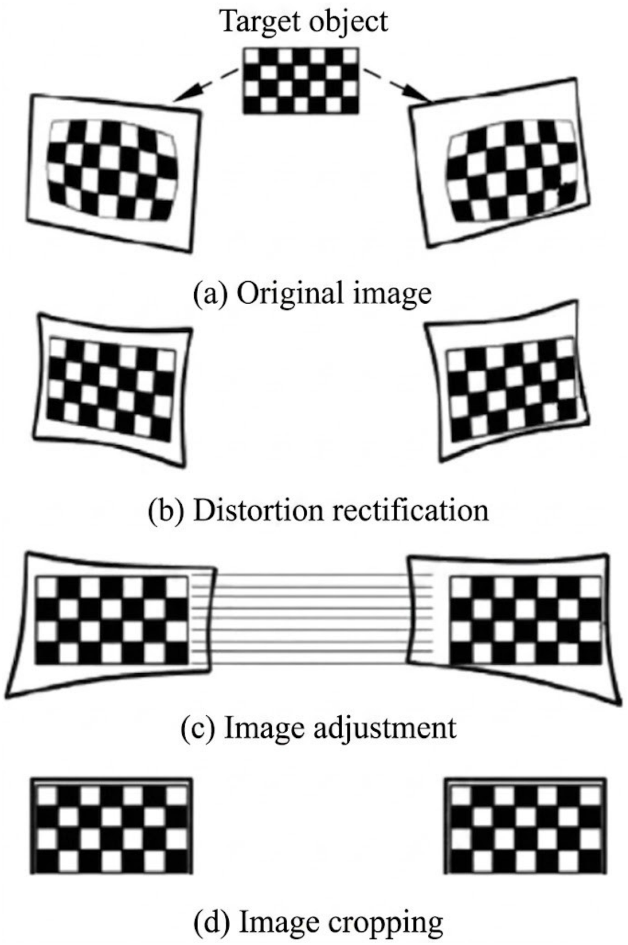
Abstract. We present a linear rectification algorithm for general, unconstrained stereo rigs. The algorithm takes the two perspective projection matrices of the original cameras, and computes a pair of rectifying projection matrices. It is compact (22-line MATLAB code) and easily reproducible. We report tests proving the correct behavior of our method, as well as the negligible decrease of the accuracy of 3D reconstruction performed from the rectified images directly.

3.3 Stereo Vision | Image Adjustment

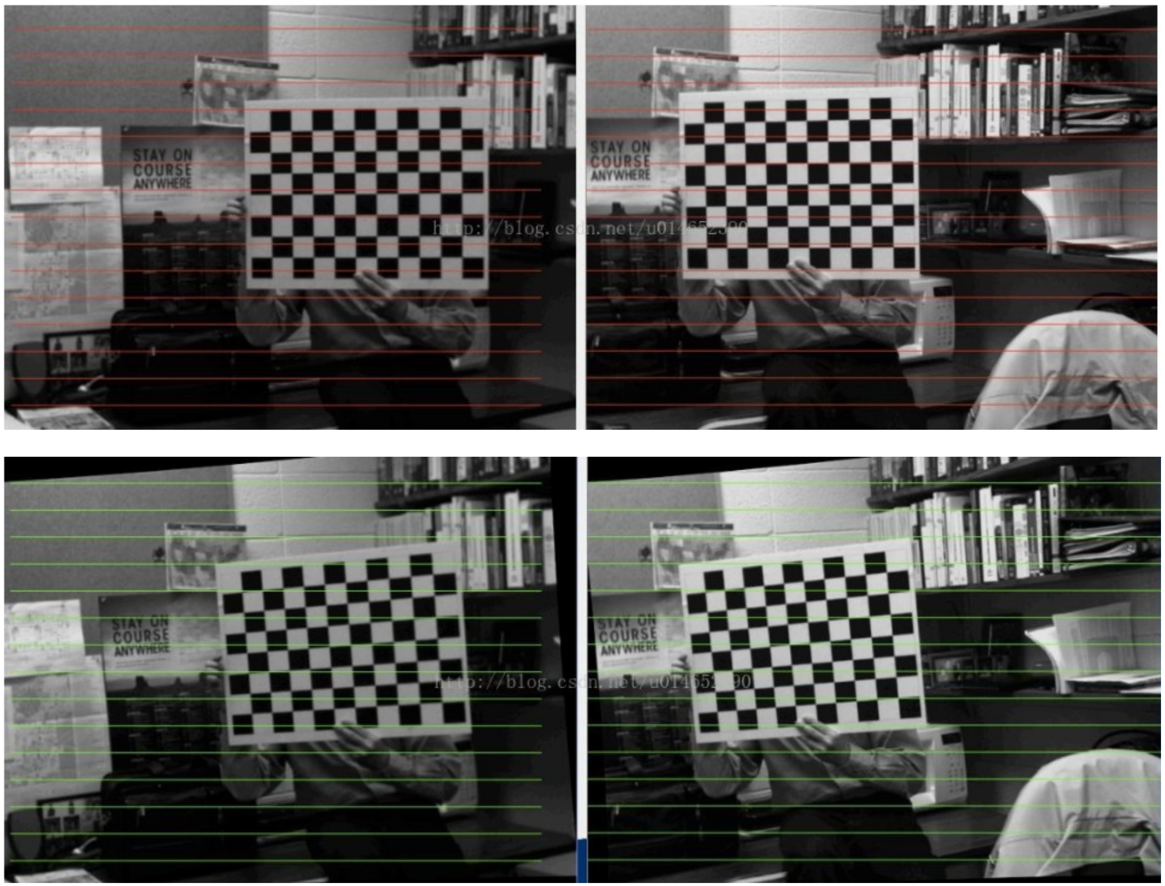
- Through camera calibration on a stereo vision system, obtain $\boldsymbol{A}_1$, $\boldsymbol{A}_2$, $\boldsymbol{R}_L$, $\boldsymbol{T}_L$, $\boldsymbol{R}_R$, $\boldsymbol{T}_R$, $\boldsymbol{R}$, $\boldsymbol{T}$
- Row alignment of the images can $\boldsymbol{R}_{rectify}$: the matrix $\boldsymbol{R}_{rectify} = \begin{bmatrix} \boldsymbol{r}_1^{\mathrm{T}} \\ \boldsymbol{r}_2^{\mathrm{T}} \\ \boldsymbol{r}_3^{\mathrm{T}} \end{bmatrix} = \begin{bmatrix} \left(\dfrac{\boldsymbol{T}}{\|\boldsymbol{T}\|}\right)^{\mathrm{T}} \\ \boldsymbol{R}_L(3,:) \times \boldsymbol{r}_1 \\ (\boldsymbol{r}_1 \times \boldsymbol{r}_2)^{\mathrm{T}} \end{bmatrix}$
- Compute the intrinsic parameter matrix of the virtual camera:
- Compute the intrinsic parameter matrix $\boldsymbol{A}_n = \dfrac{\boldsymbol{A}_1 + \boldsymbol{A}_2}{2}$
- Calculate the transformation matrices:
- Calculate the $\begin{aligned} \boldsymbol{T}_1 &= \boldsymbol{A}_n \boldsymbol{R}_{rectify} \boldsymbol{R}_L^T \boldsymbol{A}_1^{-1} \\ \boldsymbol{T}_2 &= \boldsymbol{A}_n \boldsymbol{R}_{rectify} \boldsymbol{R}_R^T \boldsymbol{A}_2^{-1} \end{aligned}$

3.3 Stereo Vision | Image Adjustment

- After the original images of the target object are captured by the left and right cameras, the image rectification process first corrects various distortions, then adjusts the left and right images to make them coplanar and row-aligned.
- For each integer coordinate in the rectified image (c), the image rectification process finds the corresponding coordinate in the corrected image (b), which is then mapped to floating-point coordinates in the original image (a). The pixel value at the floating-point coordinate is then obtained using interpolation algorithms from the surrounding integer-coordinate pixel values. This pixel value is assigned as the pixel intensity at the integer coordinate in the rectified image (c).
- Finally, the rectified left and right images typically need to be cropped (d) to extract the overlapping region used for calculating depth information.



3.3 Stereo Vision | Image Adjustment

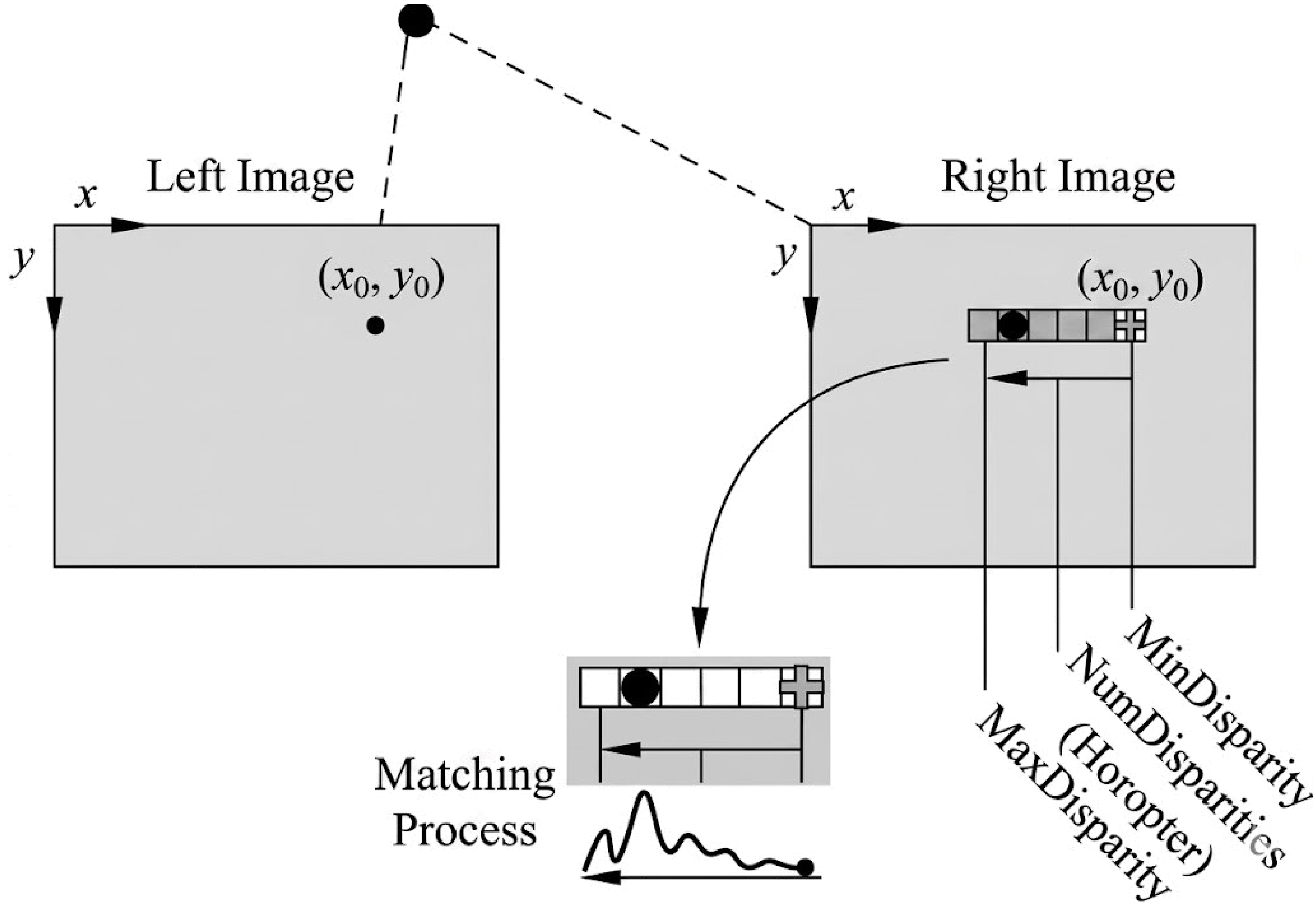


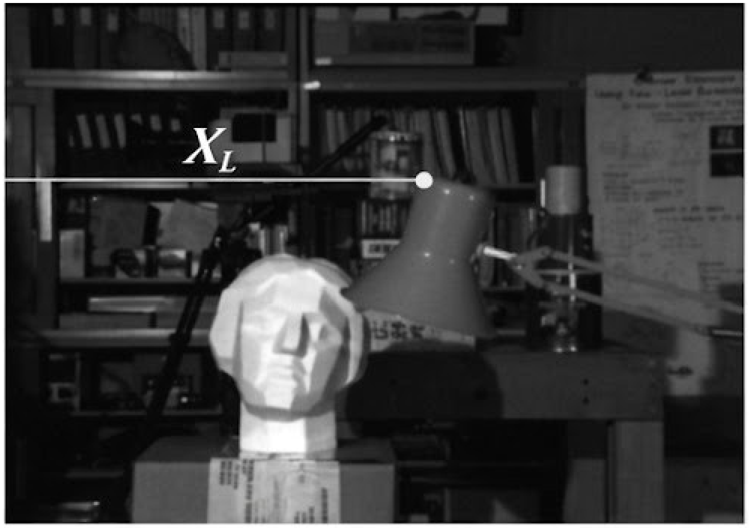
3.3 Stereo Vision | Image Adjustment

- The correspondence matching process typically outputs a disparity map for the left and right images. The disparity map takes one image of the stereo vision system (usually the left image) as a reference, and its size is the same as that of the reference image, but each element value represents the disparity of homologous points between the left and right images.
- Since the disparity between homologous points in the two images is usually small, it is generally multiplied by a factor (e.g., multiplied by 16) before display. Because in a parallel binocular vision system, the distance from a spatial point to the camera is inversely proportional to the disparity, brighter pixels (with larger grayscale values) in the disparity map indicate that the corresponding spatial point is closer to the camera; conversely, darker pixels indicate that the spatial point is farther from the camera.

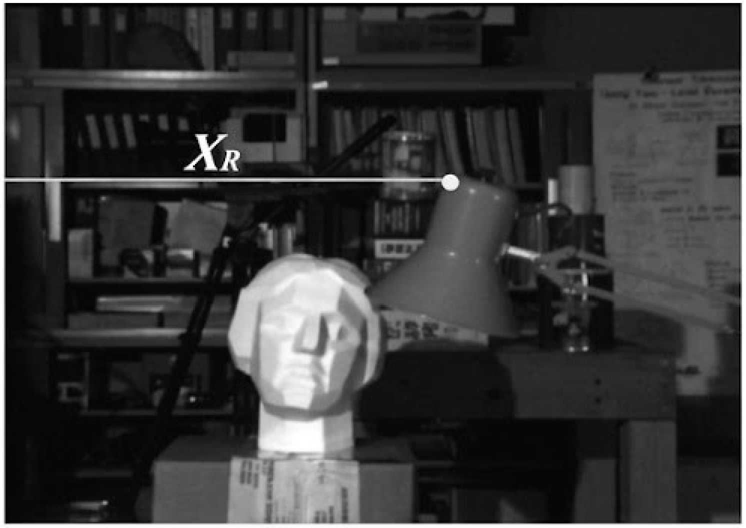
3.3 Stereo Vision | Matching and Reconstruction

- The three commonly used constraints in the process of finding corresponding pixel points are:
  - Uniqueness constraint: A given pixel in the left image cannot correspond to more than one pixel in the right image.
  - Monotonic order constraint: If a pixel is located to the left (or right) of another pixel in the left image, then its corresponding pixel in the right image should also be located to the left (or right) of the corresponding pixel of the other pixel, and vice versa. That is, the order between pixels remains consistent in both the left and right images.
  - Continuity constraint: The disparity map should change continuously across different regions in the image.

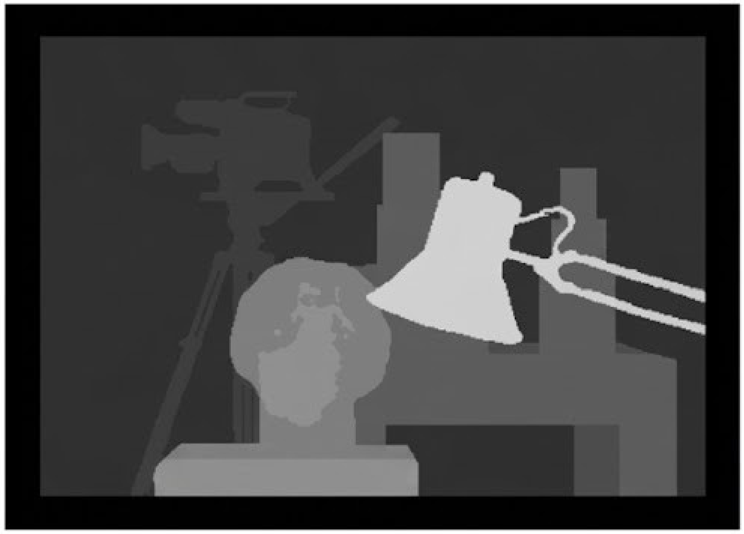

Left Image
Right Image
x
y
(x_0, y_0)
x
y
(x_0, y_0)
Matching
Process
MinDisparity
NumDisparities
(Horopter)
MaxDisparity



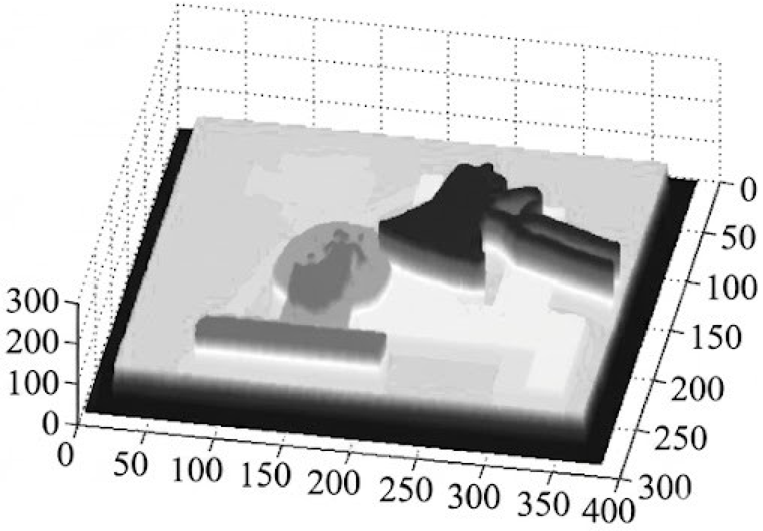
(a) Left image / Reference image



(b) Right image



(c) Disparity map 2D display



(d) Disparity map 3D display

3.3 Stereo Vision | Matching and Reconstruction

- Two common correspondence point matching algorithms: Block Matching and Semi-Global Matching.
  - Block Matching algorithms execute faster,
  - Semi-Global Matching can provide disparity maps with finer intervals and adapt to images with less detail or almost no texture.

3.3 Stereo Vision | Matching and Reconstruction

- After completing the entire corresponding point matching process, a disparity map between the left and right images can be obtained. Based on the disparity values between these corresponding points, the three-dimensional information of the target object can be further calculated. If we denote the disparity at point (x, y) in the left image as d, then: :

$$\begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \\ \boldsymbol{Z} \\ \boldsymbol{W} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -\boldsymbol{c}_x \\ 0 & 1 & 0 & -\boldsymbol{c}_y \\ 0 & 0 & 0 & f \\ 0 & 0 & \dfrac{-1}{\boldsymbol{T}_x} & \dfrac{(\boldsymbol{c}_x - \boldsymbol{c}'_x)}{\boldsymbol{T}_x} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{d} \\ \boldsymbol{1} \end{bmatrix}$$

- After obtaining $(X, Y, Z, W)^T$, the three-dimensional coordinates (X_w, Y_w, Z_w) of the target point can be calculated:

$$X_w = X/W,\ Y_w = Y/W,\ Z_w = Z/W$$